Practice workbook · Solving trig equations with a multiple or shifted angle

Solving sin, cos or tan of a single, multiple or shifted angle equal to a constant over a given interval.

13 question parts: 1 worked-example retry and 12 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-TRI-24

1. Question 8 · part a · 3 marks

2020 JAN · Paper 01

Worked-example retry · Attempt part a only.

8 (a) Solve, to the nearest integer, the equation

$$\sin(2x - 75)° = -0.515 \quad \text{for } 0 \leqslant x < 180$$

(3)

(b) Giving your solutions to one decimal place, where appropriate, solve the equation

$$2\tan y° + 5\sin y° = 0 \quad \text{for } 0 \leqslant y \leqslant 180$$

(4)

(c) Explain mathematically why there are no values of θ that satisfy the equation

$$3\cos^2\theta° - 3\sin^2\theta° + \sin\theta° + 12 = 0$$

(4)

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1. Question 8 · part a · 3 marks · Working and answer

Working and answer

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2. Question 10 · part c · 5 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

10 (a) Use the formula for $\cos(A + B)$ to show that $\cos 2A = 2\cos^2 A - 1$

(2)

(b) Show that $\cos 4A = 8\cos^4 A - 8\cos^2 A + 1$

(4)

(c) Solve the equation $\cos^2\left(\frac{\theta}{4} + \frac{\pi}{24}\right)\left[\cos^2\left(\frac{\theta}{4} + \frac{\pi}{24}\right) - 1\right] = -\frac{1}{16} \qquad 0 \leqslant \theta < 2\pi$

Give your answers in terms of π.

(5)

$$f(A) = 4\cos^4 A - 4\cos^2 A + 1$$

(d) Using calculus, find the exact value of $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} f(A)\,dA$

Give your answer in the form $a\pi - b\sqrt{c}$ where a and b are fractions in their lowest terms and c is a prime number.

(4)

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2. Question 10 · part c · 5 marks · Working and answer

Working and answer

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3. Question 7 · part b · 4 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

7 Given that

$$\frac{3+\sin^2\theta}{\cos\theta-2}=3\cos\theta$$

(a) show that $\cos\theta=-\frac{1}{2}$

(4)

(b) Hence solve the equation

$$\frac{3+\sin^2 3x}{\cos 3x-2}=3\cos 3x \quad \text{for } 0^\circ \leqslant x < 180^\circ$$

(4)

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3. Question 7 · part b · 4 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 7 continued

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(Total for Question 7 is 8 marks)

3. Question 7 · part b · 4 marks · Working and answer

Working and answer

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4. Question 10 · part a · 3 marks

2021 JUN · Paper 02

Further practice · Attempt part a only.

10 (a) Solve the equation

$$\tan x^\circ = -3 \qquad \text{for } 0 \leqslant x < 360$$

Give your solutions to the nearest whole number.

(3)

Given that

$$7\sin^2\theta + \sin\theta\cos\theta = 6$$

(b) show that

$$\tan^2\theta + \tan\theta - 6 = 0$$

(3)

(c) Hence solve the equation

$$7\sin^2 y^\circ + \sin y^\circ \cos y^\circ = 6 \qquad \text{for } 0 \leqslant y < 360$$

Give your solutions to the nearest whole number.

(4)

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4. Question 10 · part a · 3 marks · Working and answer

Working and answer

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5. Question 10 · part a · 3 marks

2021 NOV · Paper 01

Further practice · Attempt part a only.

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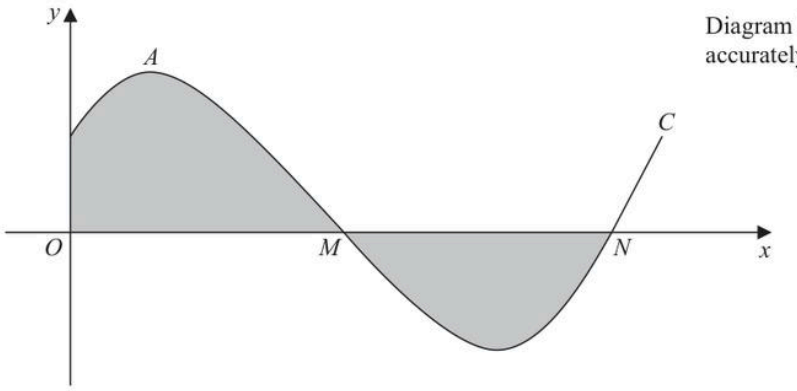


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows the curve C with equation $y = \frac{1}{2} + \sin 3x$ where $0 \leqslant x < \frac{2\pi}{3}$

The curve C crosses the x-axis at the points M and N

(a) Show that the coordinates of M are $\left(\frac{7\pi}{18}, 0\right)$ and find the coordinates of N

(3)

The curve C has a maximum at the point A

(b) Find the coordinates of A

(4)

(c) Find an equation of the tangent to C at M

Give your answer in the form $ay + b\sqrt{3}x - c\sqrt{3}\pi = 0$ where a, b and c are integers to be found.

(4)

The finite region, shown shaded in Figure 4, is bounded by the curve C, the y-axis and the part of the x-axis from O to N

(d) Use algebraic integration to find, to 3 significant figures, the total area of the shaded region.

(4)

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5. Question 10 · part a · 3 marks · Working and answer

Working and answer

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6. Question 7 · part a · 4 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

7 (a) Solve the equation

$$\cos(3x - 15)° = \frac{\sqrt{3}}{2} \quad \text{for} \quad 0 \leqslant x < 180$$

(4)

(b) Solve, giving your solutions to one decimal place where appropriate,

$$3\tan y° + 4\sin y° = 0 \quad \text{for} \quad -180 \leqslant y < 180$$

(4)

(c) Solve, giving your solutions to one decimal place where appropriate,

$$\cos\theta° = 3\sin^2\theta° - 1 \quad \text{for} \quad -180 \leqslant \theta < 180$$

(4)

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6. Question 7 · part a · 4 marks · Working and answer

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7. Question 1 · part c · 3 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part c only.

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1 Here is a formula

$$P = 3 + 2\sin\left(\frac{3\pi t}{8}\right) \qquad 0 \leqslant t \leqslant 12$$

(a) Find the exact value of P when $t = \frac{10}{3}$

(2)

(b) Find

(i) the largest value of P

(ii) the smallest value of P

(2)

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(c) Find the least value of t for which $P = 4$

(3)

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7. Question 1 · part c · 3 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 1 continued

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7. Question 1 · part c · 3 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 1 continued

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(Total for Question 1 is 7 marks)

7. Question 1 · part c · 3 marks · Working and answer

Working and answer

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8. Question 10 · part a · 3 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part a only.

10 (a) Solve the equation

$$\sin\left(x+\frac{\pi}{3}\right)=\frac{\sqrt{3}}{2} \qquad \text{for } 0 \leqslant x \leqslant 2\pi$$

Give your solutions in terms of π, where appropriate.

(3)

(b) Solve the equation

$$3\sin\theta + 5\cos\theta = 0 \qquad \text{for } -360° \leqslant \theta \leqslant 360°$$

Give your solutions to the nearest degree.

(3)

(c) Solve the equation

$$1 + \sin 2y = 2\cos^2 2y \qquad \text{for } -180° \leqslant y \leqslant 0°$$

(5)

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8. Question 10 · part a · 3 marks · Working and answer

Working and answer

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9. Question 7 · part a · 3 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

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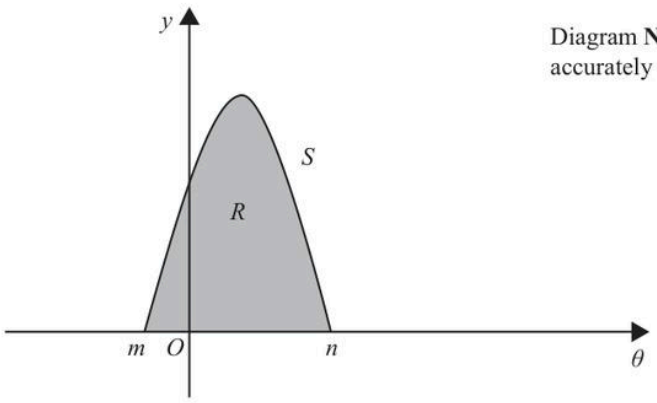


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Figure 2

Figure 2 shows part of the curve S with equation $y = \left(\cos 3\theta + \sqrt{3}\sin 3\theta\right)^{\frac{1}{2}}$

where $m \leqslant \theta \leqslant n$

The curve S meets the x-axis at the point with coordinates $(m, 0)$ and at the point with coordinates $(n, 0)$

(a) Find the exact value of m and the exact value of n

(3)

The finite region R, shown shaded in Figure 2, is bounded by the curve S, and the x-axis in the region $m \leqslant \theta \leqslant n$

The region R is rotated through 2π radians about the theta-axis.

(b) Use calculus to find the exact volume of the solid generated.

(4)

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9. Question 7 · part a · 3 marks · Working and answer

Working and answer

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10. Question 7 · part b · 3 marks

2022 JUN · Paper 02

Further practice · Attempt part b only.

7 (i) (a) Using a formula from page 2, show that

$$\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$$

(2)

Given that $\tan 2\alpha = 1$

(b) show that $\tan\alpha = a \pm \sqrt{b}$ where a and b are integers whose values need to be found.

(3)

(ii) (a) Using formulae from page 2, show that $\cos(x-30)^\circ = \sin(x+30)^\circ$ can be written as $\tan x^\circ = 1$

(4)

(b) Hence, or otherwise, solve

$$\cos(2y-30)^\circ = \sin(2y+30)^\circ \qquad \text{for } -90 < y \leqslant 90$$

(2)

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10. Question 7 · part b · 3 marks · Working and answer

Working and answer

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11. Question 11 · part c · 4 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

11 (a) Use a formula on page 2 to show that $\sin^2 A = \frac{1}{2}(1 - \cos 2A)$

(3)

(b) Show that $\sin^4 x + \cos^4 x = \frac{3 + \cos 4x}{4}$

(5)

(c) Hence solve, in degrees to one decimal place, the equation

$$8\sin^4\left(\frac{\theta}{2}\right) + 8\cos^4\left(\frac{\theta}{2}\right) = 5\sin(2\theta) + 6 \quad \text{for} \quad 0^\circ \leqslant \theta < 180^\circ$$

(4)

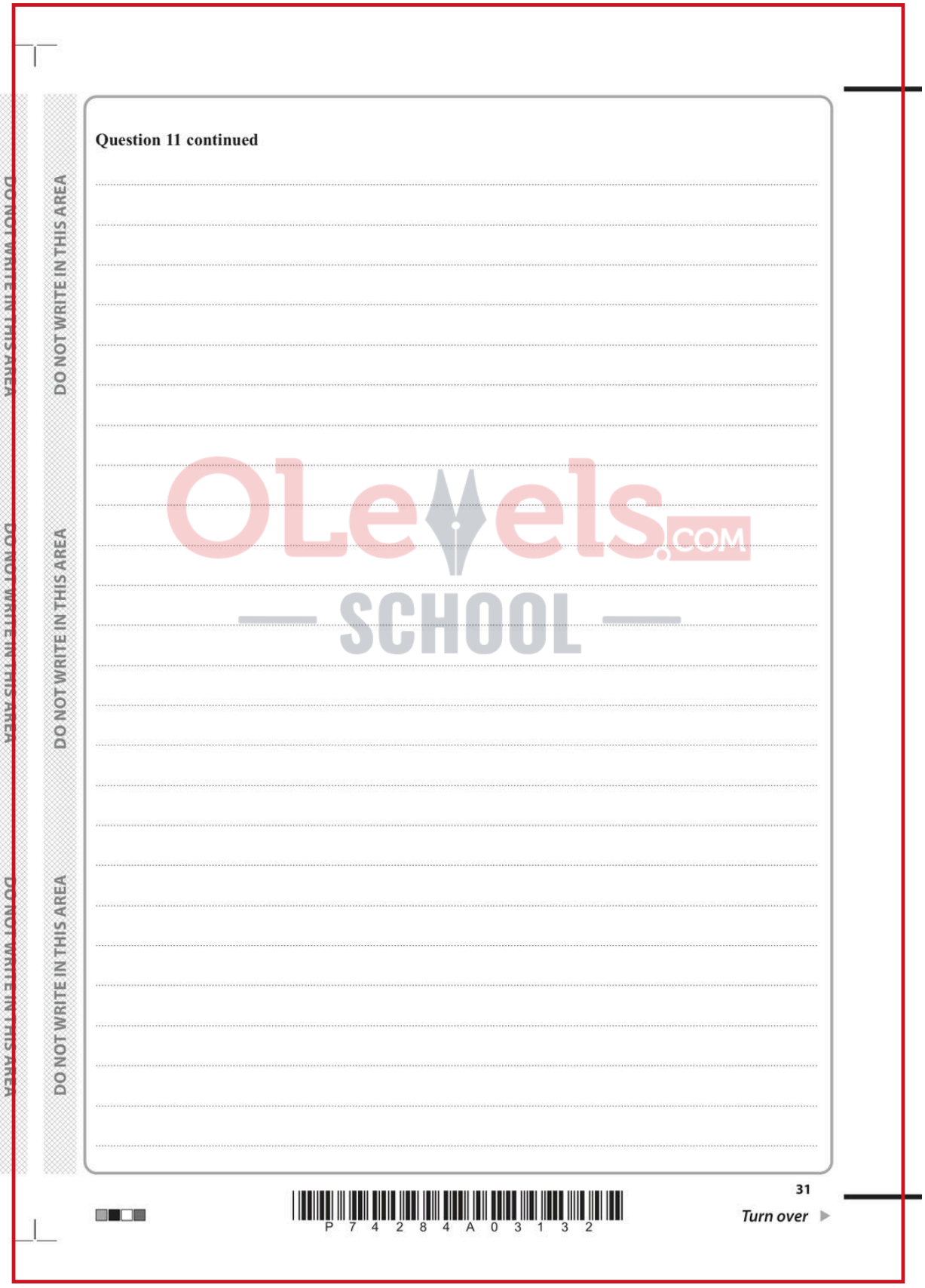
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11. Question 11 · part c · 4 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.



Question 11 continued

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11. Question 11 · part c · 4 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 11 continued

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11. Question 11 · part c · 4 marks · Working and answer

Working and answer

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12. Question 10 · part c · 6 marks

2023 NOV · Paper 01

Further practice · Attempt part c only.

10 (a) Using formulae on page 2, show that

(i) $\sin 2A = 2\sin A\cos A$

(ii) $\cos 2A = 2\cos^2 A - 1$

(3)

$$f(\theta) = \frac{2\tan\theta}{1+\tan^2\theta}$$

(b) Show that $f(\theta) = \sin 2\theta$

(4)

(c) Solve, in radians to 3 significant figures, for $-\frac{\pi}{2} \leqslant x \leqslant \frac{\pi}{2}$, the equation

$$5\tan\left(x+\frac{\pi}{6}\right) = \left[1+\tan^2\left(x+\frac{\pi}{6}\right)\right]\left[1-2\cos^2\left(x+\frac{\pi}{6}\right)\right]$$

(6)

(d) Using calculus, find the exact value of

$$\int_0^{\frac{\pi}{2}} \left(\frac{4\tan\theta}{1+\tan^2\theta} - \cos 5\theta + 2\right) d\theta$$

(4)

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12. Question 10 · part c · 6 marks · Working and answer

Working and answer

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13. Question 7 · part c · 2 marks

2022 JUN · Paper 02

Further practice · Attempt part c only.

7 (i) (a) Using a formula from page 2, show that

$$\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$$

(2)

Given that $\tan 2\alpha = 1$

(b) show that $\tan\alpha = a \pm \sqrt{b}$ where a and b are integers whose values need to be found.

(3)

(ii) (a) Using formulae from page 2, show that $\cos(x-30)^\circ = \sin(x+30)^\circ$ can be written as $\tan x^\circ = 1$

(4)

(b) Hence, or otherwise, solve

$$\cos(2y-30)^\circ = \sin(2y+30)^\circ \qquad \text{for } -90 < y \leqslant 90$$

(2)

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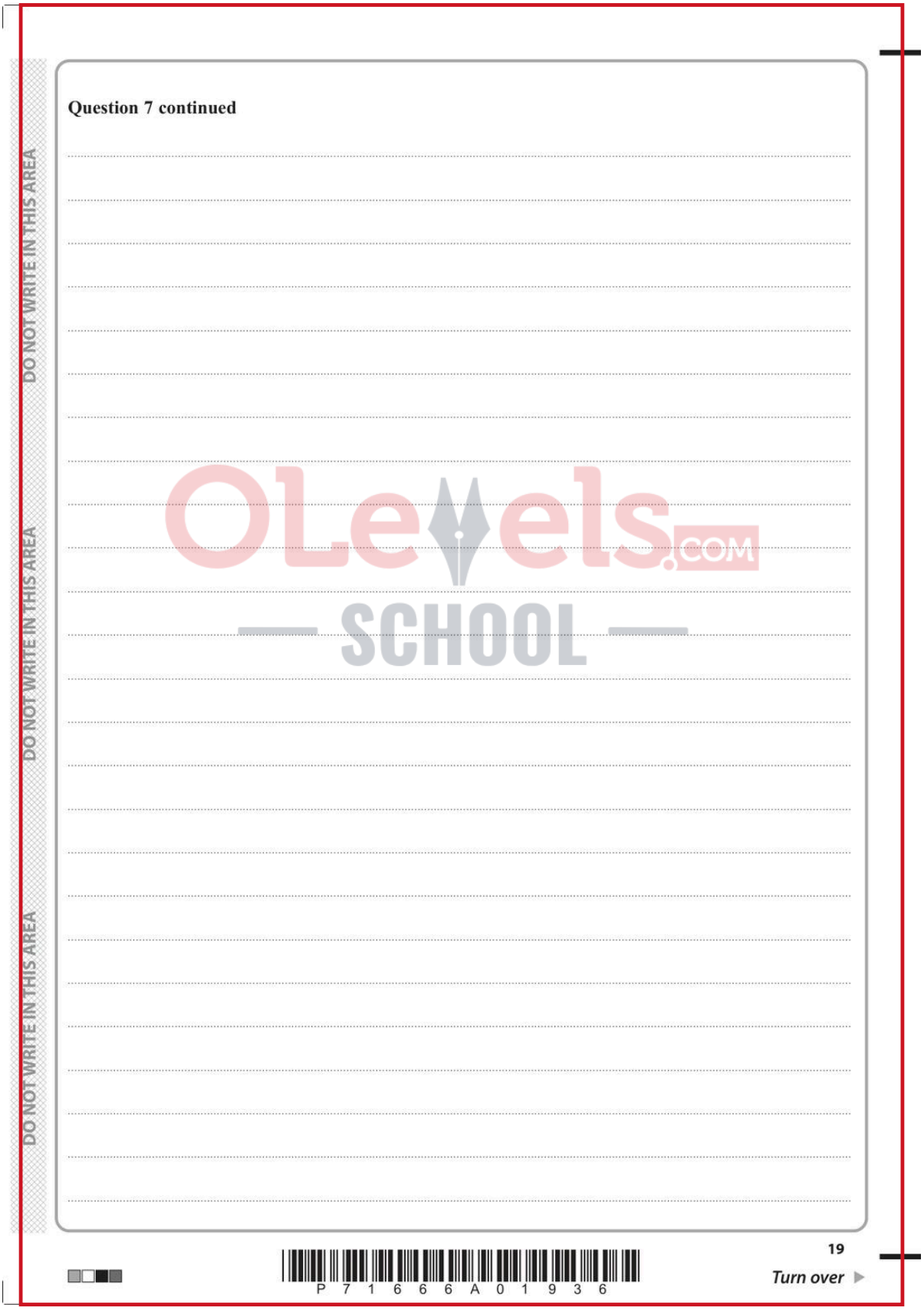
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13. Question 7 · part c · 2 marks

2022 JUN · Paper 02

Further practice · Attempt part c only.



13. Question 7 · part c · 2 marks

2022 JUN · Paper 02

Further practice · Attempt part c only.

Question 7 continued

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13. Question 7 · part c · 2 marks

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Question 7 continued

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(Total for Question 7 is 11 marks)

13. Question 7 · part c · 2 marks · Working and answer

Working and answer

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